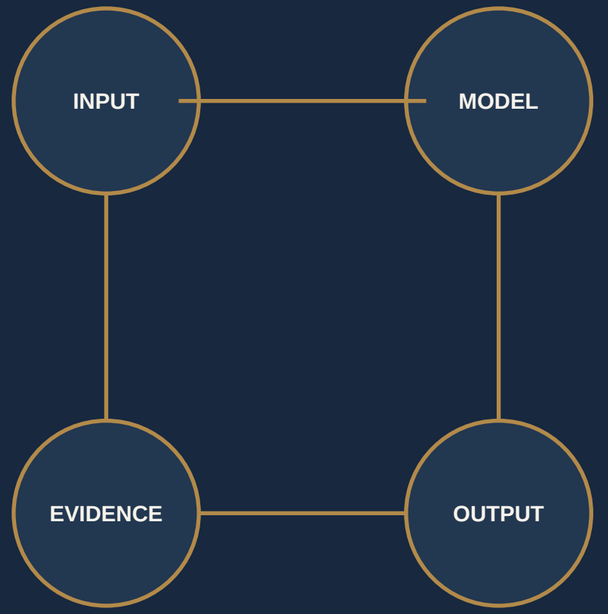
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Manipulation

AI.405 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define robotic arms.
- Apply key methods including grippers.
- Build or analyze a practical system involving grasping.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Robotic arms

02

METHODS

Grippers

03

SYSTEMS

Grasping

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Robotic arms

Robotic arms

02 · Grippers

Grippers

03 · Grasping

Grasping

04 · Force control

Force control

05 · Manipulation planning

Manipulation planning

06 · Object handling and dexterou

Object handling and dexterous manipulation

07 · Mandatory practical: Simulat

Mandatory practical: Simulate a robot arm performing pick-and-place

08 · Grasping and manipulation task

Grasping and manipulation tasks

Core ideas, made explicit

AI.405 • Distinguish definitions, assumptions and evidence

- Robotic arms
- Grippers
- Grasping
- Force control
- Manipulation planning

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.405 • Distinguish definitions, assumptions and evidence

- Object handling and dexterous manipulation
- Mandatory practical: Simulate a robot arm performing pick-and-place
- Grasping and manipulation tasks

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Robotic arms

INPUTS
Grippers

METHOD
Grasping

OUTPUT
Force control

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving robotic arms.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using robotic arms.
- Implement or simulate a baseline using grippers.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.